

Activity: Explore Cryptography in Action

Part 1. Output reads:

Invalid key length: 5 bytes

The following algorithms will be used based on the size of the key:

16 bytes = AES-128

24 bytes = AES-192

32 bytes = AES-256

The input key length shows 15, the output 159

Part 2. #It does not have SHA-256 as an option, but SHA2 has it in its subset, so I dragged that one over. When I add a forward slash to the end the entire hash changes completely. I do not see the algorithm.

Part 3. That website's issuer is:

Common Name (CN)	WE2
Organization (O)	Google Trust Services
Organizational Unit (OU)	<Not Part Of Certificate>

-Key length (-unknown)

-Encryption Algorithm is: Elliptic Curve Public Key

-Validity Period is from Monday, July 20, 2026 at 1:06:16 PM, and expires on Monday, October 12, 2026 at 1:06:15 PM.

The encryption algorithm used in CyberChef activity is AES encrypt. Key length is important because it affects how difficult it is for an attacker to crack encryption.

A hash is different from encryption because hashing is one-way, while encryption is reversible with the correct key.

A digital certificate helps prove that a website is legitimate and can be trusted.